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Effects of different long-day photoperiods on somatic growth and gonadal development in Nile tilapia (*Oreochromis niloticus* L.)

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Abstract

Long-day photoperiods are considered as an effective managerial tool in manipulating somatic growth and reproduction in a number of fish species. Effects of three different artificial long-day photoperiods on somatic growth and gonadal development of Nile tilapia (*Oreochromis niloticus* L.) were investigated in this study. Swim-up fry with a mean initial weight of 0.06 g were exposed to 24L:0D, 20L:4D and 18L:6D artificial photoperiods and ambient light regime (control) for 24 weeks at 27.0±1 °C. The effect of photoperiodic manipulation was only detectable and statistically meaningful during fingerling stage. Long-day photoperiods resulted in significantly higher mean final weights and specific growth rates (SGR) than natural light regime. The highest mean final weight (24.94±0.45 g) and SGR (3.46±0.03% day^{-1}) were obtained under 24L:0D photoperiod. Mean female gonadosomatic index (GSI) and mean oocyte size were significantly lower in fish maintained under continuous light regime (24L:0D) than those of 20L:4D, 18L:6D treatments and control. The highest gonadosomatic indices were recorded in control female and males. Mean oocyte diameter in fish exposed to continuous light was measured as 1.05±0.06 mm with the bulk of the oocytes (60.0%) in pre-vitellogenic stage (≤1.20 mm). On the contrary, oocyte size and size distribution of oocytes in 20L:4D, 18L:6D photoperiod groups and control were indicating a more advanced oocyte development stage, i.e. vitellogenic (1.2–1.8 mm) and post-vitellogenic stages (>2.1–2.4 mm). Basically, results obtained support the idea that continuous artificial lighting may be influential on enhancing somatic growth and delaying gonadal development in Nile tilapia during fingerling stage.

Keywords: Nile tilapia; Long-day photoperiods; Somatic growth; Gonadal development

1. Introduction

With increasing consumer's acceptance and demand for tilapia fillets, the tilapia farming industry has been steadily expanding in the last few years, with 85 countries producing 1.526 million metric tons in 2003 (Fiorillo, 2003; Fitzsimmons, 2004). However, tendency of females to mature and reproduce early at small size before reaching market size remains one of the main biological constraints for tilapia farming (Longalong et al., 1999; Lorenzen, 2000; Lutz et al., 2003). The maturation process involves significant changes in teleosts including inhibition of growth mechanism (Gines et al., 2003). Tilapia species tend to sacrifice

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growth to maintain reproductive capacity (Coward and Bromage, 1999a) and have also been reported to exhibit tremendous plasticity in growth and maturation (Turner and Robinson, 2000). In culture conditions, females that do not mature at an early stage often follow the growth patterns of males (Longalong et al., 1999). In physiological terms, early sexual maturity results in reduced growth and consequently poor feed conversion performances. In economic terms, this phenomenon means higher feed costs and lower profitability for tilapia enterprises.

Different approach and techniques have been employed to overcome this biological constraint in tilapia farming including hormonal sex reversal, hybridization, manual sexing, use of predators, cage culture, high stocking density, sterilization, selection and use of YY male broodstock (Hörstgen-Schwark and Langholz, 1998; Longalong et al., 1999; Lorenzen, 2000). Photoperiodic manipulation is emerging as an acceptable approach of practical application for regulating physiological functions in fish farming. Photoperiod techniques have been found to be influential on many physiological functions in fish species such as growth, reproduction and gonadal maturation, and are now widely used in aquaculture to alter spawning season, manipulate maturation and stimulate growth (Bromage, 1995; Boeuf and Le Bail, 1999; Purchase et al., 2000; Bromage et al., 2001; Randall et al., 2001; Rodriguez et al., 2001; Biswas and Takeuchi, 2002; Gines et al., 2003). In recent years, long photoperiod regimes have been used in Atlantic salmon (*Salmo salar*), rainbow trout (*Oncorhynchus mykiss*), Atlantic halibut (*Hippoglossus hippoglossus*), sea bream (*Sparus aurata*) and sea bass (*Dicentrarchus labrax*) to enhance growth and/or to delay sexual maturity (Boeuf and Le Bail, 1999; Oppedal et al., 1999; Jonassen et al., 2000; Simensen et al., 2000; Kissil et al., 2001; Randall et al., 2001; Rodriguez et al., 2001; Gines et al., 2004). Growth-stimulating long photoperiod treatments either improve food-processing efficiency directly or suppress sexual maturation thus redirecting energy from gonadal development to somatic growth (Boeuf and Le Bail, 1999; Gines et al., 2004).

Reproduction of many sub-tropical and tropical species is also cued by photoperiod (Bromage et al., 2001). The effects of photoperiodic manipulation on reproduction of tilapia species are not well understood (Campos-Mendoza et al., 2004). However, studies carried out by Ridha et al. (1998), Ridha and Cruz (2000) and Campos-Mendoza et al. (2004) in recent years demonstrate that reproductive performance and seed production in tilapia species (*Oreochromis niloticus* and *Oreochromis spilurus*) are influenced by photoperiod manipulation and that photoperiod could be used as an powerful environmental tool for broodstock management in tilapia farming (Campos-Mendoza et al., 2004). The most recent study carried out by Biswas et al. (2005) suggest that reproductive activities in Nile tilapia (*O. niloticus*) (230–340 g) can be arrested by photoperiod manipulation and fish exposed to 6L:6D photoperiod could not successfully spawn after three to four spawning cycle. Moreover, photoperiodic manipulations have been found to influence metabolic rate and growth of Nile tilapia through energy conservation and stimulated food intake (Biswas et al., 2002; Biswas and Takeuchi, 2002; Biswas and Takeuchi, 2003; El-Sayed and Kawanna, 2004). However, the impact of long-day photoperiodic regimes on somatic growth-sexual maturation axis in Nile tilapia from early life stages is not well studied. This study was therefore conducted to investigate the possibilities of delaying gonadal development and improving somatic growth in Nile tilapia by exerting constant long-day artificial photoperiods throughout fry and fingerling stages.

2. Materials and methods

2.1. Experimental fish

This experiment was conducted with 800 Nile tilapia swim-up fries (0.06 g initial mean weight), belonging to the same batch obtained from broodstock kept in Aquaculture Research Unit of Faculty of Fisheries, University of Mersin (Turkey).

2.2. Experimental setup

To assess and compare the pattern of somatic growth and gonadal development under long-day and natural photoperiods, fish were subjected to 24L:0D, 20L:4D and 18L:6L artificial photoperiods and natural light–dark cycle (control) (36°47′N–34°38′E). The experiment was conducted indoor in Aquaculture Research Unit of Faculty of Fisheries, University of Mersin using slop-bottomed fiberglass tanks (100 × 100 × 50 cm) with a central standpipe and an outer sleeve pipe, draining from bottom to facilitate self-cleaning and waste removal. Eight tanks were arranged in four duplicate groups, six allocated to three artificial photoperiod regimes and two to control. The initial water volume in each experimental tank was 0.2 m^3. As fish grew, water volume was gradually increased to 0.35 m^3. To exclude external source of light, photoperiod groups (2 tanks/group) were placed in separate compartments made of a cubic structured metal frame and black colored PVC sheets. Control tanks were left uncovered. Light in each photoperiod tank was provided by two fluorescent lamps (36 W) suspended 100 cm above the water surface. In 24L:0D photoperiod, regime lights were continuously on, the other two photoperiod regimes (20L:4D and 18L:6D) were achieved using 24-h timers (Kengo, 230 v-50 Hz). Light intensity was measured daily at water surface (at center) by a digital Lux Meter (Digital Instrument LX-101) and was constant at around 800 lx throughout the experiment in artificial photoperiod groups. Ambient light intensities in control tanks were measured as 750–850 lx at 09:00 h and as 80–100 lx at 16.00 h. Natural light/dark cycles ranged approximately from 14.5L:09.5D at the beginning of the experiment to 09L:15D at the end of the study.

The water temperature at each experimental tank was kept constant at 27.0±1 °C by inserting thermostatically controlled 300 W heaters (Tetratec HT300). A central blower continuously aerated the tanks. Mean values of dissolved oxygen (Oxiguard oxygenmeter), unionized ammonia, NH_3 (Orbeco-Hellige, 975 MP test kit) and pH (Hanna pHmeter, HI 8314) throughout the experiment were measured as 7.30±0.20 mg/l, 0.74±0.04 mg/l and 8.20±0.04 mg/l, respectively.

2.3. Experimental procedures

Each experimental tank was stocked with 100 swim-up fries. To avoid differences in mean initial body weight among replicates and experimental groups, mean initial weight of fish in each tank was arranged to be similar (0.06 g/fry). Due to small size and fragility of fry, the initial mean weights were determined by group weighing (100 fries/group) as described by Coward and Bromage (1999a). Fish were allowed to acclimatize to experimental conditions for 24 h before any interference.

Fish were then fed a commercial feed (500, 800 and 1200 μm crumble) containing 52% proteins and 14% fat. Literature on optimal daily feeding rates for tilapia species is highly controversial. Therefore; daily feeding rates (% BW/day) were determined based on findings and recommendations of different researchers (Rakocy, 1989; De Silva and Anderson, 1995; Morris and Mische, 1999; Al Hafedh, 1999; Coward and Bromage, 1999a; El-Sayed, 2002) and experimental conditions (water renewal, dissolved oxygen). Fish were fed at a rate of 15.0% (28 days), 10.0% (28 days), 5.0% (28 days), 4.0% (14 days), 3.0% (56 days) and 2.5% (14 days) of body weight per day for 168 days. Daily rations were adjusted every 2 weeks according to fish biomass in each tank. The diet was offered in two meals at 09:00 h and 16:00 h.

Tanks were siphoned daily to remove faeces and uneaten feed. Twenty-five percent (25.0%) of water in each tank was renewed daily with dechlorinated tap water at 27 °C.

This experiment was carried out for 24 weeks. Fish were weighed at 2-week intervals (days: 14, 28, 42, 56, 70, 84, 98, 112, 126, 140, 154 and 168). Fish were weighed using an electronic balance (Shimadzu, EB-620SU±0.01 g sensitivity). To minimize damage to fragile fry, fish were counted and group weighed in a pre-weighed beaker of water for the first two weighing on days 14 and 28, similar to the initial weighing. From third weighing (day 42) until the end of the experiment, all fish in each experimental tank were netted, weighed individually and biweekly group mean recorded. At the end of 24 weeks, all fish in experimental tanks were harvested and individually weighed to determine group mean final weights. Randomly selected 50 fish from each tank was used for studying gonadal development.

At the end of the experiment, growth rates were measured in terms of "specific growth rate" (SGR) as following:

$$\text{SGR }(\%\ \text{day}^{-1}) = [L_{\text{n}}(W_{\text{f}}) - L_n(W_{\text{i}})/T] \times 100$$

Where W_{f} was the individual final weight at day 168 (T) and W_{i} was mean initial group weight (0.06 g) since fish were only group weighed at the beginning of the experiment. Individual SGR figures were used to determine group means.

For each weighing period, feed conversion ratio (FCR) was computed by the following formula (Bailey et al., 2003):

$$\text{FCR} = \text{Feed delivered (kg) to group} \\ /\text{Live biomass gain (kg) of that group}$$

Biweekly data were used to compute group mean FCRs for the entire experiment.

Fish were categorized as fry (<5.0 g) or fingerling (5.0–30.0 g) based on their mean body weights according to Chapman (2000).

Gonadal development was assessed in terms of gonadosomatic index (GSI), mean oocyte size (mm), oocyte size distribution and development stages. GSI was computed in both female and males as following (Biswas and Takeuchi, 2003):

$$\text{GSI }(\%) = [\text{wet weight of gonad (g)} \\ /\text{wet total body weight (g)}] \times 100$$

Oocyte diameter was measured according to Coward and Bromage (1999b) as:

$$\text{Oocyte diameter (mm)} = [\text{long axis length} + \text{short axis length}]/2$$

Oocytes were classified as pre-vitellogenic, vitellogenic, post-vitellogenic and post-vitellogenic mature oocyte based on oocyte diameter according to Gunasekera and Lam (1997).

2.4. Statistical analysis

Factorial analysis of variance was used to investigate interactions and to compare the effects of tanks, time (weighing days) and photoperiod on mean body weights. To cancel out the effect of tanks on mean body weights of fish, tanks were included in the model as a block. Tukey's multiple comparison test was used for determination of significance differences between treatments. For SGR, FCR, GSI and egg diameter, the statistical significance between treatments was evaluated by one-way analysis of variance (ANOVA) and Tukey's procedure was used for multiple comparisons. All analysis was carried out by *STATISTICA* (version 6.0) packet program. Type I error was accepted as 0.05. Average values are given as mean±S.E.M.

3. Results

3.1. Growth performance

Data and illustrations on growth patterns of Nile tilapia exposed to three long-day artificial photoperiod (24L:0D, 20L:4D and 18L:6D) and ambient light regime (control) are presented in Tables 1 and 2 and Fig. 1. At the end of 24-week experiment, fish exposed to artificial long-day photoperiods reached significantly higher mean final weights and SGRs than control ($p<0.05$). The highest mean final weight (24.94±0.45 g) was reached in fish maintained under continuous light regime (24L:0D). Mean final weights of fish exposed to 20L:4D and 18L:6D photoperiods were measured as 19.84±0.43 and 19.06±0.44 g, respectively, and found to be significantly lower than that of 24L:0D group ($p<0.05$). Mean final weight in control fish was measured as 14.68±0.43 g.

While factorial analysis of variance revealed significant interactions between the effect of photoperiod on body weight and weighing days ($p<0.001$), no significant differences in mean body weights were detected

Table 1
Growth pattern of Nile tilapia under different photoperiod treatments

Time (day)	Mean body weight (g)			
	24L:0D	20L:4D	18L:6D	Control
0 (initial)	0.06	0.06	0.06	0.06
14	0.43	0.39	0.43	0.37
28	1.14	1.07	1.18	1.12
42	1.99±0.80	1.70±0.80	2.32±1.12	1.86±0.80
56	2.66±0.80	2.00±0.84	3.09±0.56	2.40±0.80
70	4.05±0.44	3.23±0.43	4.89±0.43	3.54±0.42
84	5.36±0.44[a]	4.19±0.43[b]	6.39±0.43[ab]	4.09±0.42[ab]
98	6.92±0.44[ab]	5.19±0.43[a]	8.11±0.43[b]	4.64±0.42[ab]
112	8.89±0.44[a]	6.14±0.43[b]	8.98±0.43[a]	5.18±0.42[b]
126	12.72±0.44[a]	9.13±0.43[b]	11.58±0.43[c]	7.19±0.43[d]
140	18.29±0.45[a]	12.99±0.43[b]	14.79±0.43[b]	10.00±0.43[c]
154	22.98±0.45[a]	16.69±0.43[b]	16.65±0.44[b]	12.32±0.43[c]
168 (final)	24.94±0.45[a]	19.84±0.43[b]	19.06±0.44[b]	14.68±0.43[c]

*Due to group weighing and unavailability of individual body weight data, first two weighing periods were not included in ANOVA. Values are means±S.E.M.; means in the same row with different superscript are significantly different ($P<0.05$) from each other.

either within artificial photoperiod groups themselves or between photoperiod groups and control until day 70th or fry stage (<5.0 g). Significant differences in mean body weights of photoperiod groups and control were only detectable during the last 6 weeks (days 126–168) of the experiment or during fingerling stage (5.0–30.0 g). Throughout this stage, mean body weight of fish at constant light regime (24L:0D) was significantly different and higher than that of other two photoperiod regimes (20L:4D and 18L:6D) and control ($p<0.05$). Moreover, mean body weights of fish exposed to 20L:4D and 18L:6D photoperiod groups were also found to be significantly different from that of control during this development stage (Table 1). No systematic mortalities were recorded throughout the experiment. Few incidences of accidental mortalities were recorded where fish had jumped out of tanks. Five such incidences were recorded in one of the tanks in 24L:0D group and two in 18L:6D photoperiod group.

Mean SGR and FCR figures are presented in Table 2. The highest mean SGR (3.46±0.03% day^{-1}) was recorded in fish maintained under 24-h continuous light. This figure was found to be statistically significant that mean SGR figures computed for 20L:4D and 16L:6D photoperiod groups and control ($p<0.05$). The lowest mean SGR was observed (3.19±0.02% day^{-1}) in control, being significantly lower than those of three photoperiod groups ($p<0.05$). However, the difference between mean SGR values obtained under 20L:4D (3.35±0.02% day^{-1}) and 18L:6D (3.29±0.03%

Table 2
Specific growth rate (SGR) and feed conversion ratios (FCR) of fish exposed to different photoperiods

	24L:0D	20L:4D	18L:6D	Control
SGR (% day^{-1})	3.46±0.03^{a}	3.35±0.02^{b}	3.29±0.03^{b}	3.19±0.02^{c}
FCR	1.77±0.32	2.10±0.55	2.06±0.32	2.37±0.45

Values are means±S.E.M.; means in the same row with different superscript are significantly different ($P<0.05$) from each other.

day^{-1}) photoperiod regimes were not found to be statistically significant ($p>0.05$). The best mean FCR figure (1.77±0.32) was also obtained under continuous lighting (24L:0D). Mean FCR figures for 20L:4D and 18L:6D photoperiod regimes were computed as 2.1±0.54 and 2.06±0.32, respectively. The poorest mean FCR (2.37±0.45) was recorded in control. However, the difference between mean FCR values computed under different treatments was not found to be statistically significant ($p>0.05$) (Table 2).

3.2. Gonadal development

Mean values of gonadosomatic index for female and males, mean oocyte size and distributions are presented in Table 3. Significant differences were detected between mean GSIs of female and male fish maintained under 24L:0D photoperiod and those of control ($p<005$). Females subjected to 24L:0D photoperiod regime had the lowest GSIs, measured as 1.79 ±0.24%. GSIs of male fish were similar in all three photoperiod treatments, but were significantly lower from that of control. The highest GSIs were observed in female and males of control, measured as 5.14 ±0.66% and 1.37±0.30%, respectively (Table 3).

Sixty percent (60.0%) of oocytes in ovarian of fish maintained under 24L:0D photoperiod regime were ≤1.2 mm in size and in pre-vitellogenic stage. While the portion of pre-vitellogenic oocytes (≤1.2 mm) in ovaries of females in 20L:4D, 18L:6D and control were recorded as 26.0%, 40.0% and 20.0%, respectively. Great portion of oocytes in 20L:4D, 18L:6D treatments and control were >1.2 mm in size. Eighty percent (80.0%) of oocytes in control group was >1.2 mm in size. The smallest mean oocyte size (1.05±0.07 mm) was therefore recorded in ovarian of female fish exposed to 24-h continuous light regime and the largest (1.61±0.12 mm) in the females of control. Mean oocyte diameters in fish exposed to 20L:4D and 18L:6D treatments were measured as 1.56±0.12 and 1.35±0.20 mm, respectively. Only the difference between mean oocyte size of 24L:0D treatment and that of control was found to be statistically significant ($p<0.05$). While post-vitellogenic and mature oocytes were recorded in females kept under 20L:4D, 18L:6D photoperiods and in control, no oocyte in post-vitellogenic or matures stages were recorded in ovarian of fish under 24L:0D regime (Table 3).

4. Discussion

Mean final weights and growth rates (SGR) of fish exposed to different artificial photoperiods and ambient light–dark cycle in this study (Tables 1 and 2) reveal that growth in Nile Tilapia is enhanced under constant long-day artificial photoperiods (24L:0D,

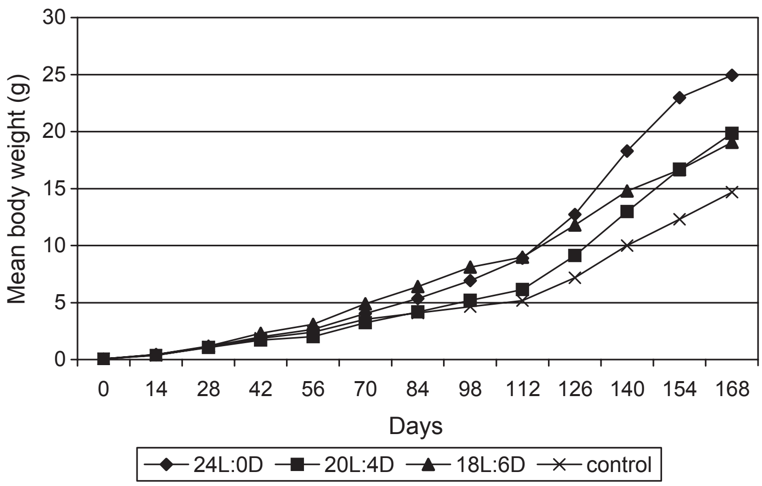

Fig.1. Growth performance of Nile tilapia under different photoperiod regimes.

Table 3
Gonadosomatic indices (GSI), diameter and percentage size distribution of oocytes in fish exposed to different photoperiods

	24L:0D	20L:4D	18L:6D	control
Mean GSI (%)				
Female	1.79 ± 0.24^{a}	3.31 ± 0.45^{ab}	3.43 ± 0.87^{ab}	5.14 ± 0.66^{b}
Male	0.59 ± 0.08^{a}	0.56 ± 0.07^{a}	0.53 ± 0.05^{a}	1.37 ± 0.30^{b}
Oocyte				
Mean diameter (mm)	1.05 ± 0.07^{a}	1.56 ± 0.12^{ab}	1.35 ± 0.20^{ab}	1.61 ± 0.12^{b}
Size distribution (%)				
≤1.2 mm, pre-vitellogenic	60.0	26.0	40.0	20.0
>1.2–1.5 mm, vitellogenic	29.0	15.0	13.0	18.0
>1.5–1.8 mm, vitellogenic	8.0	18.0	12.0	27.0
>1.8–2.1 mm, vitellogenic	3.0	20.0	18.0	22.0
>2.1–2.4 mm, post-vitellogenic	–	11.0	11.0	8.0
>2.4 mm, mature oocyte	–	10.0	6.0	5.0

Values are means ± S.E.M.; means in the same row with different superscript are significantly different ($P < 0.05$) from each other.

20L:4D and 18L:6D) when compared to natural light regime. Long-day photoperiods have been reported to stimulate growth in a number of fish species (Boeuf and Le Bail, 1999; Randall et al., 2001). Biswas and Takeuchi (2003) have similarly reported clear effect of photoperiod manipulation on growth in Nile tilapia. Improved growth in Nile tilapia under longer day photoperiods is further supported by Biswas and Takeuchi (2002) and Biswas et al. (2002) findings who have demonstrated that metabolic rate and energy loss were negatively correlated with light phase and Nile tilapia could conserve more energy under longer light phases.

Furthermore; fish maintained under continuous artificial lighting (24L:0D) in this study attained significantly higher final weights and SGRs compared to other two long-day photoperiods regimes (Tables 1 and 2). The significantly higher mean final weight and growth rate obtained under 24L:0D treatment mean that continuous light regime is more effective in enhancing somatic growth in Nile tilapia than 20L:4D and 18L:6D light cycles. Growth enhancing effect of continuous artificial lighting has also been reported for rainbow trout (*O. mykiss*) by Randall et al. (2001).

Improved growth in Nile tilapia under 24-h continuous light has also been reported by El-Sayed and Kawanna (2004) who documented best weight gain and SGR in Nile tilapia fry, maintained under 24 and 18 h long light periods without any significant difference between two treatments. El-Sayed and Kawanna (2004) have, however, not found a significant difference in growth performances of Nile tilapia under different photoperiods during fingerling stage. They have concluded that photoperiodic manipulation of growth in Nile tilapia depends on development stage and fry are more sensitive (0.02–1.24 g) to photoperiod than fingerlings (2.33 g–51.35 g.). In our study, the effect of long-day photoperiods on growth of Nile tilapia was not detectable until day 70th or during fry stage when mean weights were less than 5.0 g. Influence of long-day photoperiods was statistically meaningful during the last 6 weeks of the experiment or so called fingerling stage (5.0–30.0 g). Throughout this stage, mean weight of fish exposed to 24L:0D photoperiod was significantly different from those of 20L:4D, 18L:6D and control. Moreover, during this period, mean weights of fish exposed to other two long-day artificial photoperiods (20L:4D and 18L:6D) were also significantly different from that of control (Table 1) showing the overall impact of long-day photoperiods on growth of Nile tilapia in fingerling stage. Lourenco et al. (1998) have also documented the growth enhancing effect of photoperiod on fingerling stage of Nile tilapia (El-Sayed and Kawanna, 2004). The controversy between our finding and El-Sayed and Kawanna (2004) in terms of impact of photoperiod manipulation on growth of Nile tilapia during different life stages, i.e. fry or fingerling, could be attributed to employment of different statistical methods or experimental conditions such as light intensities applied since along with photoperiod light intensity also appear to play a major role in most species during very young stage (Boeuf and Le Bail, 1999). The light intensity applied by El-Sayed and Kawanna (2004) was kept constant at 2500 lx throughout their experiment.

Different responses of Nile tilapia to photoperiodic manipulation during fry and fingerling stages recorded in this study or by other researches are further supported by findings of Boeuf and Le Bail (1999) who have reported that receptivity of fish to light profoundly changes according to species and development status.

Growth-stimulating affect of long-day photoperiods is attributed to either improvement of food conversion efficiency or to suppression of sexual maturation and redirection of energy from gonadal development to somatic growth by different researchers (Boeuf and Le Bail, 1999; Gines et al., 2003; Gines et al., 2004). Improved somatic growth in fish maintained under long-day photoperiods and 24L:0D treatment in particular in this experiment may also be explained by improved feed conversion ratios (FCR) or delayed gonadal development.

As shown in Table 2, even though statistically insignificant the mean FCR values computed for artificial photoperiod groups were relatively better than that of control with the lowest FCR (1.77±0.32) obtained under 24L:0D regime. Improved FCR in Nile tilapia fry under long-day light regimes (24L:0D and 18L:6D) has been also reported by El-Sayed and Kawanna (2004). Higher weight gains under long-day photoperiods and particularly 24L:0D treatment in this may also be explained by delayed gonadal development. Suppression of sexual maturity and redirection of food energy towards somatic growth rather than gonadal development by exerting long-day photoperiods have been documented in a number of fish species (Randall et al., 2001; Gines et al., 2003; Gines et al., 2004; Biswas et al., 2005). Biswas and Takeuchi (2003) have also reported delayed gonad maturation in Nile tilapias with higher specific growth rates. As illustrated in Table 3, mean GSIs and oocyte sizes were lower in fish maintained under long-day photoperiods than control fish. However, this pattern was more evident and statistically significant in fish exposed to 24L:0D photoperiod. Both female and male fish maintained under continuous light possessed significantly lower GSIs than those exposed to natural light regime (control). Mean oocyte size was also significantly smaller in fish exposed to 24L:0D regime. Determination of exact size at first maturation by using GSI is difficult in *O. niloticus* since they do not have a distinct spawning period and different stages of oocyte development and related GSI levels are found in females (de Graaf et al., 1999). Typically, mouth-breeding tilapia exhibit GSIs of between 4.6% and 10.2% (Coward and Bromage, 2000). de Graaf et al. (1999) have used a GSI of 2.9% as a major indicator for first maturity in Nile tilapia and have reported mature females as small as 12.7 g with a GSI of 5.3%. Even though occurrence of first sexual maturation in control fish could not be verified precisely by mean GSI values, based on oocyte size distributions and development (Table 3), it would be possible to state that primarily the control fish and secondly those exposed to 20L:4D and 18L:6D photoperiods in the present experiment were at least in a more advanced stage of gonad development than fish exposed to continuous lighting. Mean oocyte diameter in fish exposed to continuous light was measured as 1.05±0.06 mm with the bulk of the oocytes (60.0%) in pre-vitellogenic stage (≤1.20 mm). While in control and in 20L:4L and 18L:6D photoperiod groups, mean oocyte diameters and size distribution of oocytes were indicative of a more advanced oocyte development stage, i.e. vitellogenic stage (1.2–1.8 mm) as defined by Gunasekera and Lam (1997). Contrary to control, 20L:4D and 18L:6D treatments, no post-vitellogenic or mature oocytes were observed in ovarian of females maintained under 24L:0D photoperiod. Oocyte size distributions and statistically significant difference between mean GSI values and oocyte sizes in 24L:0D group and control also supports the assumption that gonadal development was delayed under continuous light. Desynchronized spawning and a reduction in the proportion of fish attaining sexual maturity by using continuous artificial lighting have also been reported for *O. mykiss* (Randall et al., 2001).

The main objective of this study was to assess the impact of long-day photoperiodic regimes on somatic growth-sexual maturation axis in Nile tilapia rather than exploring the growth potential of Nile tilapia under experimental conditions employed. Therefore, rather than considering merely the growth figures, patterns of somatic growth and gonadal development under different photoperiods are important here. In this context, data obtained and statistical assessments conducted in this study show that artificial continuous light (24L:0D) regime seems to enhance somatic growth in Nile tilapia during fingerling stage and to delay gonadal development. However, the impact of continuous artificial light regime on somatic growth–gonadal development axis need to be further supported by detailed studies on alternation in hormonal and physiological activities.

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References

Al Hafedh, Y.S., 1999. Effects of dietary protein on growth and body composition of Nile tilapia, *Oreochromis niloticus* L. Aquaculture Research 30, 385–393.

Bailey, J., Alanara, A., Crampton, V., 2003. Do delivery rate and pellet size affect growth rate in Atlantic salmon (*Salmo salar* L.) raised under semi-commercial farming conditions? Aquaculture 224, 79–88.

Biswas, A.K., Takeuchi, T., 2002. Effects of different photoperiod cycles on metabolic rate and energy loss of both fed and unfed adult tilapia *Orochromis niloticus*: part II. Fisheries Science 68, 543–553.

Biswas, A.K., Takeuchi, T., 2003. Effects of photoperiod and feeding interval on food intake and growth of Nile tilapia *Orochromis niloticus* L. Fisheries Science 69, 1010–1016.

Biswas, A.K., Endo, M., Takeuchi, T., 2002. Effects of different photoperiod cycles on metabolic rate and energy loss of both fed and unfed young tilapia *Orochromis niloticus*: part I. Fisheries Science 68, 465–477.

Biswas, A.K., Morita, T., Yoshizaki, G., Maita, M., Takeuchi, T., 2005. Control of reproduction in Nile tilapia *Oreochromis niloticus* (L.) by photoperiod manipulation. Aquaculture 243, 229–239.

Boeuf, G., Le Bail, P.L., 1999. Does light have an influence on fish growth? Aquaculture 177, 129–152.

Bromage, N.R., 1995. Broodstock management and seed quality-general considerations. In: Bromage, N.R., Roberts, R.J. (Eds.), Broodstock Management and Egg and Larval Quality. Blackwell Sciences, London, p. 424.

Bromage, N., Porter, M., Randall, C., 2001. Environmental regulation of maturation in farmed finfish with special reference to the role of photoperiod and melatonin. Aquaculture 197, 63–99.

Campos-Mendoza, A., McAndrew, B.J., Coward, K., Bromage, N., 2004. Reproductive response of Nile tilapia (*Oreochromis niloticus*) to photoperiod manipulation; effects on spawning periodicity, fecundity and egg size. Aquaculture 231, 299–313.

Chapman, F.A., 2000. Culture of hybrid Tilapia: a reference profile. Cooperative Extension Service, Institute of Food and Agricultural Sciences University of Florida, Circular 1051.

Coward, K., Bromage, N.R., 1999a. Spawning frequency, fecundity, egg size and ovarian histology in groups of Tilapia zilli maintained upon two distinct food ration sizes from first-feeding to sexual maturity. Aquatic Living Resources 12, 11–22.

Coward, K., Bromage, N.R., 1999b. Spawning periodicity, fecundity and egg size in laboratory-held stocks of substrate-spawning tilapiine, *Tilapia zilli* (Gervais). Aquaculture 171, 251–267.

Coward, K., Bromage, N.R., 2000. Reproductive physiology of female tilapia broodstock. Reviews in Fish Biology and Fisheries 10, 1–25.

de Graaf, G.J., Galemoni, F., Huisman, E.A., 1999. Reproductive biology of pond reared Nile tilapia, *Oreochromis niloticus* L. Aquaculture Research 30, 25–33.

De Silva, S.S., Anderson, T.A., 1995. Fish Nutrition in Aquaculture. Chapman and Hall, London. 319 pp.

El-Sayed, A.M., 2002. Effects of stocking density and feeding levels on growth and feed efficiency of Nile tilapia (*Oreochromis niloticus* L.). Aquaculture Research 33, 621–626.

El-Sayed, A.F., Kawanna, M., 2004. Effects of photoperiod on the performance of farmed Nile tilapia *Oreochromis niloticus*: I. Growth, feed utilization efficiency and survival of fry and fingerlings. Aquaculture 231, 393–402.

Fiorillo, J., 2003. Should you climb aboard the tilapia bandwagon? IntraFish Newspaper, December. Intrafish Media A.S., Norway.

Fitzsimmons, K., 2004. Development of new products and markets for the global tilapia trade. In: Bolivar, R., Mair, Fitzsimmonas, G. (Eds.), Proceedings of Sixth International Symposium on Tilapia in Aquaculture, Manila, Philippines, pp. 624–633.

Gines, R., Afonso, J.M., Argüello, A., Zamorano, M.J., Lopez, J.L., 2003. Growth in adult gilthead sea bream (*Sparus aurata* L.) as a result of interference in sexual maturation by different photoperiod regimes. Aquaculture Research 34, 73–83.

Gines, R., Afonso, J.M., Argüello, A., Zamorano, M.J., Lopez, J.L., 2004. The effects of long-day photoperiod on growth, body composition and skin colour in immature gilthead sea bream (*Sparus aurata* L.). Aquaculture Research 35, 1207–1212.

Gunasekera, R.M., Lam, T.J., 1997. Influence of dietary protein level on ovarian recrudescence in Nile tilapia, *Oreochromis niloticus* (L.). Aquaculture 149, 57–69.

Hörstgen-Schwark, G., Langholz, H.-J., 1998. Prospects of selecting for late maturity in tilapia (*Oreochromis niloticus*): III. A selection experiment under laboratory conditions. Aquaculture 167, 123–133.

Jonassen, T.M., Imsland, A.K., Kadowaki, S., Stefansson, S.O., 2000. Interaction of temperature and photoperiod on growth of Atlantic halibut *Hippoglossus hippoglossus*. Aquaculture Research 31, 219–227.

Kissil, G.W., Lupatsch, I., Elizur, A., Zohar, Y., 2001. Long photoperiod delayed spawning and increased somatic growth in gilthead sea bream (*Sparus aurata*). Aquaculture 200 (3–4), 363–379.

Longalong, F.M., Eknath, A.E., Bentsen, H.B., 1999. Response to bi-directional selection for frequency of early maturing females in Nile tilapia (*Oreochromis niloticus*). Aquaculture 178, 13–25.

Lorenzen, K., 2000. Population dynamics and management. In: Beveridge, M.C.M., MCAndrew, B.J. (Eds.), Tilapias: Biology and Exploitation. Kluwer Academic Publishers, Netherlands, p. 505.

Lutz, C.G., Sambidi, P., Harrison, R.W., 2003. Tilapia industry profile. Agricultural Marketing Resource Center, Ag MRC, Iowa State University, USA. (http://test.agmrc.org/aquaculture/profiles/tilapia profiles.pdf). September 2003.

Morris, J.E., Mische, C.C., 1999. Tilapia Culture in Iowa. Iowa State University, University Extension, USA.

Oppedal, F., Taranger, G.L., Juell, J-E., Hansen, T., 1999. Growth, osmoregulation and sexual maturation of under yearling Atlantic salmon (*Salmo salar*) exposed to different intensities of continuous light in sea cages. Aquaculture Research 30 (7), 491–499.

Purchase, C.F., Boyce, D.L., Brown, J.A., 2000. Growth and survival of juvenile yellowtail flounder (*Pleuronectes ferrugineus* Storer) under different photoperiods. Aquaculture Research 31, 547–552.

Rakocy, J.E., 1989. Tank culture of tilapia. Southern Regional Aquaculture Center, SRAC, USA. publication no 282.

Ridha, M.T., Cruz, E.M., 2000. Effect of light intensity and photoperiod on Mile tilapia *Oreochromis niloticus* seed production. Aquaculture Research 31, 609–717.

Ridha, M.T., Cruz, E.M., Al-Ameeri, A.A., Al-Ahmed, A.A., 1998. Effects of controlling temperature and light duration on seed production in tilapia, *Oreochromis spilurus* (Günter). Aquaculture Research 29, 403–410.

Randall, C., North, B., Futter, W., Porter, M., Bromage, N., 2001. Photoperiod effects on reproduction and growth in rainbow trout. Trout News 32, 12–16.

Rodriguez, L., Zanuy, S., Carrillo, M., 2001. Influence of day length on the age at first maturity and somatic growth in male sea bass (*Dicentrarchus labrax* L.). Aquaculture 196, 159–175.

Simensen, L.M., Jonassen, T.M., Imsland, A.K., Stefansson, S.O., 2000. Photoperiod regulation of growth of juvenile Atlantic halibut (*Hippoglossus hippoglossus*). Aquaculture 190 (1–2), 119–128.

Turner, G.F., Robinson, R.L., 2000. Reproductive biology, mating systems and parental care. In: Beveridge, M.C.M., MCAndrew, B. J. (Eds.), Tilapias: Biology and Exploitation. Kluwer Academic Publishers, Netherlands, p. 505.